

$$\begin{array}{c}
 y = e^x \\
 \downarrow \\
 \text{Reverse} \\
 \downarrow \\
 y = \ln x \\
 \downarrow \text{same} \\
 e^y = x
 \end{array}$$

$$\begin{array}{c}
 y = \ln(x^x) \\
 \downarrow \\
 y^{\ln(x^x)} = x^x
 \end{array}$$

$$\begin{array}{l}
 \ln(e^3) = 3 \\
 \Rightarrow e^3 = e^3 \Rightarrow 3
 \end{array}$$